

CASE

Moderna: Democratizing Artificial Intelligence

“A technology company that happens to do biology.”

Stéphane Bancel, CEO

~6,000

employees

100,000+

staff at rivals

10

approvals in 3 years



April 2023: the forcing event

Samsung engineers leak source code into public ChatGPT.

Moderna's exposure is worse.
Not just drug research — the medical records of clinical trial volunteers.

“It was a burning platform for us. People were using it to solve real business needs, and we didn't want to take that away — but doing nothing on the security front would put us at risk.”

Adrian Masson, AI Transformation Lead

Two doors: ban it · or replace it.

What they did instead of banning it

May 2023 — public ChatGPT cut off. **mChat ships in two weeks** on OpenAI's API. Nothing trains the model.

Then the adoption playbook — the transferable part:

- 1 Prompt competition** 180+ entries. Top prize: a trip to meet OpenAI's founders. Winners chosen by peer reactions.
- 2 AI Academy** Training, town halls, employees demoing to each other.
- 3 8 weeks × 1 hour** Senior leadership training, plus office hours. They became the champions.

POLL 1 · PREDICT

How many custom GPTs do you think Moderna's employees had built by December 2024?

- A About 50
- B About 200
- C About 1,400
- D About 10,000

Wooclap results

Participation is graded. Answer is not



By December 2024: 1,400+ GPTs

Almost all built by employees. Almost none by the AI team.

#	GPT	Messages	Users
1	Self-Review GPT	12,751	885
2	U.S. Benefits Assistant	7,887	1,215
3	Red Ribbon Visuals	7,884	593
12	CatGPT — turns your photo into a cat	1,370	170
18	Zeus, the GPT of Moderna GPTs	446	84

“Generative AI cuts through this entire process because there's no coding.” — Masson

The case says 1,400. Zeus — the GPT built to find the other GPTs — says approximately 2,000.

Source: HBS 625-070, Exhibit 1 — Top 20 Custom GPTs by messages, December 2024

POLL 2 · PREDICT

Which was Moderna's single most-used custom GPT?

- A** A regulatory bot for FDA submissions
- B** A contract-drafting tool for the legal team
- C** A tool that helps you write your own performance review
- D** A tool that turns your photo into a cat

Wooclap results

Participation is graded. Answer is not



Two GPTs that made leadership stop

SELF-REVIEW GPT

The most-used tool in the company.
Helps employees write their own performance reviews.

“Augmenting — and to some extent replacing — a critical process in developing employees. Maybe there are some processes that should be kept off-limits?”

Tracey Franklin, Chief People & Digital Technology Officer

DoseID

Recommends drug dosing for clinical trials.
Cuts the decision from a month to a fortnight.

Does this need regulatory approval? Clinical trials are tightly regulated.

Stéphane Bancel, CEO

One is high-stakes because of regulators and patients. The other because of people and judgment. Neither is a technology failure.

POLL 3 · YOUR VIEW

Your future employer gives you an AI tool that writes your performance self-review. Do you use it?

- A** Yes, definitely
- B** Yes, but I'd heavily rewrite it
- C** No — I'd write it myself
- D** Depends whether my manager knows I used it

Wooclap results

No correct answer. Participation is graded. Answer is not

What could go wrong

“The risk is we have a hallucination and we send something to regulators that is incorrect. If I decentralize, I'm assuming there are enough guardrails from my quality and legal teams.” — Bancel

And the unglamorous failures:

- Four** competing travel-expense GPTs — none built by the travel-expense team
- Stale** nobody knows how to retire a GPT once it is out of date
- Orphaned** no clear owner when the employee who built it leaves

The counter-example is in the case. Other pharma companies simply banned ChatGPT. Samsung's leak happened 20 days after they lifted their own ban.

December 2024

“Our approach so far has been to empower employees and let a thousand flowers bloom — but as we reach that number of custom GPTs, we may need to adjust our strategy.”

Brad Miller, CIO

THREE QUESTIONS ON THE TABLE

- 1 Should Moderna take centralised ownership of GPTs?
- 2 Is the current governance sufficient — or does it need more oversight?
- 3 Does Moderna need to tell regulators that GPTs are now inside core processes?

POLL 4 · YOUR VIEW

Should Moderna take centralised control of employee-built GPTs?

A Yes — centralise all of them

B No — keep it open

C Only the high-risk ones

D Not sure

Wooclap results

No correct answer. Anonymous to the room; recorded against each student.

Questions

10 minutes — anything about the case, the company, or the decision.

GROUP DISCUSSION

7 min

At your table

There are no wrong answers. Pick one person to report back.

- 1 Moderna's most-used AI tool helps employees write their own performance reviews. You have probably used AI on a university assignment or a personal statement. Where is the line between AI helping you say what you meant, and AI replacing the thinking you were supposed to do? Is it the same line in both cases?
- 2 Moderna has four different travel-expense GPTs and none were built by the travel-expense team. If you were a new employee on day one, which would you use — and how would you even know which to trust?
- 3 One GPT recommends drug dosing for clinical trials. Another turns your photo into a cat. Both were built the same way, by employees, with no coding. What should be different about how a company treats those two — and who decides?

